Mod-48 Page Theory — Clean Spec V1

A compact, math-clean specification of the page decomposition, bit fields, and induced operators.

1) Setup

- **Pages.** For $p \in \mathbb{Z}_{\geq 0}$, define the page

$$P_p := \{48p, 48p+1, \dots, 48p+47\}.$$

- **8-bit field.** For $n \in \mathbb{Z}_{\geq 0}$, let $F(n) \in \{0,1\}^8$ be the least-significant 8-bit vector of n . Write components $F_k(n) \in \{0,1\}$ for bit k with $k=0$ the LSB.
- **Periodicity.** $F(n+256) = F(n)$ for all n . Consequently, any construction depending only on $F(n)$ is 256-periodic.
- **LCM scale.** Page step is 48 and field period is 256. Hence the joint repeat scale is

$$L := \operatorname{lcm}(48, 256) = 768, \quad 48 \mid L,\ 256 \mid L.$$

2) Page types and orbit length

Define the page-offset $r_p := 48p \bmod 256$. Because $\gcd(48, 256) = 16$, the map $p \mapsto r_p$ has orbit length $256/16 = 16$. Therefore:

Proposition 2.1 (Sixteen page types). The array $\{F(n) : n \in P_p\}$ depends only on $p \bmod 16$. Equivalently, there are exactly 16 distinct page types, repeating with period 16 in p .

Consequence. Specifying a page type requires at most 4 bits.

3) Pagewise XOR invariant

For a finite set $S \subset \mathbb{Z}_{\geq 0}$, write the bitwise XOR

$$\bigoplus_{n \in S} F(n) := \left(\bigoplus_{n \in S} F_0(n), \dots, \bigoplus_{n \in S} F_7(n) \right).$$

Theorem 3.1 (Zero XOR on each page). For every p ,

$$\bigoplus_{n \in P_p} F(n) = (0,0,0,0,0,0,0,0).$$

Proof. Partition P_p into three consecutive blocks of length 16. For $k \leq 3$, bit k has period 2^{k+1} dividing 16, so each 16-block contains exactly 2^k ones, an even number. For $k \geq 4$, bit k is constant on each 16-block (since its period is ≥ 32), hence contributes either 0 or 16 ones per block, again even. Summing three even counts yields even parity for every bit. ∎

4) Resonance functional

Choose a fixed weight vector $\alpha \in \mathbb{R}^8$ independent of p and n . We adopt the balanced, normalized choice

$$\alpha := \frac{1}{\sqrt{24}}\,(-3,-1,-1,-1,\ 1,1,1,3)^\top.$$

Define the resonance

$$R(n) := \alpha^\top F(n) \in \mathbb{R}.$$

Basic properties. 1. **Periodicity:** $R(n+256) = R(n)$. 2. **Range:** $R(n) \in \left[-\frac{6}{\sqrt{24}}, \frac{6}{\sqrt{24}}\right] = \left[-\frac{3}{\sqrt{6}}, \frac{3}{\sqrt{6}}\right]$. 3. **Zero mean over a 256-cycle:** $\sum_{n=0}^{255} R(n) = 0$ because $\sum_k \alpha_k = 0$ and each bit is 1 exactly half the time over a full cycle.

4.1 Flux across a page

Define the pagewise flux

$$\Phi(p) := \sum_{n\in P_p} \big(R(n+1) - R(n)\big) = R(48p+48) - R(48p).$$

Theorem 4.2 (Flux telescoping and 16-page cancellation). For all p ,

$$\sum_{q=0}^{15} \Phi(p+q) = R(48p+768) - R(48p) = 0.$$

In particular, the net flux over any consecutive block of 16 pages is zero.

Remark. $\Phi(p) = 0$ holds exactly when $R(48p+48) = R(48p)$; this need not occur for every p .

5) Metrics on indices

Let $p(n) := \lfloor n/48 \rfloor$ be the page index of n . - **Page metric:** $d_P(n,m) := |p(n) - p(m)| \in \mathbb{Z}_{\ge 0}$. - **Resonance metric:** $d_R(n,m) := |R(n) - R(m)| \in \mathbb{R}_{\ge 0}$. - **Product metric:** For any $\gamma > 0$,

$$d_\gamma(n,m) := \sqrt{d_P(n,m)^2 + \gamma^2\, d_R(n,m)^2}\,.$$

Proposition 5.1. d_P and d_R are metrics. Hence d_γ is a metric on $\mathbb{Z}_{\ge 0}$.

Sketch. Nonnegativity and symmetry are clear. Identity of indiscernibles follows from equality of page indices and resonance values. Triangle inequality for d_γ follows from Minkowski since each component obeys a triangle inequality. ∎

6) A periodic weighted line and its spectrum

Consider the infinite line graph on $\mathbb{Z}$ with edge weights

$$w_n := \begin{cases}\beta, & n \equiv 47 \pmod{48} \text{ (page boundary)}, \\ 1, & \text{otherwise},\end{cases} \qquad \beta \in (0,1].$$

Let L be the normalized Laplacian of this weighted graph. The environment is 48-periodic.

Facts. 1. The graph is connected for all $\beta > 0$. The spectrum of L on $\ell^2(\mathbb{Z})$ has band structure touching 0; there is no positive spectral gap on the infinite line. 2. On a finite weighted cycle C_N with $N = 48m$ and the same w_n repeated, the algebraic connectivity $\lambda_1(C_N)$ satisfies

$$\lambda_1(C_N) = \Theta_\beta\left(\frac{1}{N^2}\right).$$

In particular, there exist constants $0 < c_1(\beta) \le c_2(\beta)$ such that

$$\frac{c_1(\beta)}{N^2} \le \lambda_1(C_N) \le \frac{c_2(\beta)}{N^2}.$$

3. For the lazy random walk associated with C_N , the total-variation mixing time obeys

$$t_{\text{mix}}(\varepsilon) \le C(\beta)\, N^2 \, \log \tfrac{1}{\varepsilon}.$$

Notes. Item 1 follows from standard Floquet analysis for periodic 1D media. Items 2–3 follow from Poincaré and Cheeger-type inequalities for weighted cycles; constants depend only on β .

7) Crossing penalties and optimization primitive

Let E be the edge set of $\mathbb{Z}$ with weights w_n as above. Define a cost functional for a walk $\pi = (n_0, \dots, n_T)$:

$$\mathcal{C}_\tau(\pi) := \sum_{t=0}^{T-1} \left(\mathbf{1}_{\{n_t \not\equiv 47\}} + \tau \mathbf{1}_{\{n_t \equiv 47\}}\right), \qquad \tau \ge 0.$$

Let $\Pi_{s\to t}$ be all nearest-neighbor paths from s to t . The τ -optimal cost is

$$L_\tau(s,t) := \min_{\pi \in \Pi_{s\to t}} \mathcal{C}_\tau(\pi).$$

Proposition 7.1 (Monotone penalty effect). If $0 \le \tau_1 \le \tau_2$ then $L_{\tau_1}(s,t) \le L_{\tau_2}(s,t)$. Moreover, any minimizer for τ_2 uses no more page-boundary edges than a minimizer for τ_1 .

Interpretation. Increasing τ discourages page crossings in any shortest-path computation, without asserting runtime formulas.

8) Concrete data at the LCM scale

- **Superperiod.** $R(n+768) = R(n)$, hence $\Phi(p+16) = \Phi(p)$ and the 16-page net flux is identically zero.
- **Counts per superperiod.** A 768-block contains 16 full pages and 3 full 256-cycles of F .

9) Minimal implementation sketch

Given α above: 1. Precompute $F(n)$ for $n = 0, \ldots, 255$. 2. Tabulate $R(n) = \alpha^\top F(n)$ over one cycle. 3. For any n , query by $R(n \bmod 256)$. For any page p , compute $\Phi(p) = R(48p + 48 \bmod 256) - R(48p \bmod 256)$ and note $\sum_{q=0}^{15} \Phi(p+q) = 0$.

This specification contains only definitions and results that hold under the stated constructions, with proofs or standard references implied for spectral statements on periodic weighted cycles.